

PROJECT SCOPE DOCUMENT

InventraX — Inventory & Asset Management Platform

Project Name	InventraX Desktop Application
Version	2.0.0
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1. Executive Summary

InventraX is a standalone desktop application developed by TechFusion Repairs LLC to provide small and medium-sized businesses with a professional, offline-capable platform for managing inventory and physical assets. The application is built using Python and PyQt5, stores all data locally in a SQLite database, and requires no internet connection or subscription service to operate.

The goal of this project is to deliver a polished, production-ready tool that replaces manual spreadsheet tracking with a purpose-built system featuring real-time dashboards, barcode scanning, automated backups, multi-format reporting, and a fully themeable interface.

2. Project Objectives

- Provide a single executable desktop application requiring no installation of databases or server software
- Replace manual Excel/spreadsheet-based inventory tracking with a structured, searchable, database-backed system
- Enable asset assignment, transfer, return, and maintenance tracking with full history logs
- Support barcode and QR code generation and scanning via USB wedge scanners and webcam
- Deliver automated SQLite backup, rotating log files, and multi-format export (Excel, CSV, PDF)
- Provide a customizable UI with 7 built-in color themes and adjustable font sizing
- Package and distribute the application as a single Windows .exe file via PyInstaller

3. Scope of Work

3.1 In Scope

The following features and deliverables are included in the v2.0.0 release of InventraX:

Feature	Description	Status
Dashboard Tab	Live stat cards, mini bar chart, popular items, getting started guide	Included
Inventory Management	Full CRUD, click-to-edit, sortable/filterable table, color-coded rows	Included
Item Detail Editor	Double-click popup with all fields: serial, warranty, notes	Included
Asset Management	Assignment, status tracking, notes, department, location	Included

Feature	Description	Status
Asset History	Full event log: assigned, returned, transferred, status change	Included
Maintenance Log	Repair/inspection records with cost tracking and next-due dates	Included
Barcode / QR Generation	QR code creation with inline preview dialog	Included
USB Scanner Support	Keyboard-wedge scanner input via dedicated dialog tab	Included
Camera Scanner	Live webcam barcode/QR scanning (requires opencv + pyzbar)	Included
Reports Tab	Text report + bar/pie/line charts in app	Included
CSV Export	Inventory and assets exported as .csv files	Included
Excel Export	Multi-sheet workbook: inventory, low stock, OOS, top selling, assets	Included
PDF Export	Professional report with styled tables (requires reportlab)	Included
SQLite Database	Persistent local database with WAL mode, foreign keys, audit log	Included
Auto-Save on Close	Inventory and assets saved to DB automatically on app exit	Included
Auto-Backup	Timestamped DB backup created on every clean close, 10-file rotation	Included
User Preferences	Theme, font size, settings persisted to user_prefs.json	Included
7 Color Themes	Teal Dark, Midnight Blue, Purple Haze, Crimson Night, Amber Forge, Arctic Light, Forest Green	Included
Keyboard Shortcuts	12 shortcuts: Ctrl+S, Ctrl+N, Ctrl+F, Ctrl+P, F5, and more	Included
Rotating Log Files	logs/inventrax.log — 5 MB × 3 file rotation	Included
Crash Handler	Uncaught exceptions logged and shown in error dialog	Included
Splash Screen	Branded loading screen shown during startup	Included
.EXE Packaging	Single-file Windows executable via PyInstaller	Included
Excel Import	Upload .xlsx inventory file via File menu	Included
Print Support	Direct print via QPrinter	Optional
Cloud Sync	Cloud storage or remote database integration	Excluded
Multi-user / Network	Networked shared database or user accounts	Excluded
Web / Mobile Version	Browser-based or mobile app version	Excluded

3.2 Out of Scope

- Cloud-based storage, syncing, or remote access
- Multi-user access control, user roles, or login authentication
- Web application or mobile (iOS/Android) version
- Integration with third-party ERP, POS, or accounting software
- Custom hardware or barcode scanner procurement
- Ongoing maintenance, support, or feature development beyond v2.0.0 unless contracted separately

4. Technical Architecture

4.1 Technology Stack

Component	Technology
Language	Python 3.10+
UI Framework	PyQt5 (Qt5 via Python bindings)
Database	SQLite 3 — local file, no server required
Charts	Matplotlib (Qt5Agg backend)
Data Processing	pandas, openpyxl
QR / Barcode	qrcode[pil], Pillow
Camera Scanning	opencv-python, pyzbar
PDF Export	reportlab
Packaging	PyInstaller (single .exe)
Logging	Python logging — RotatingFileHandler

4.2 File Structure

Path	Purpose
Main.py	Entry point — logging, DB init, splash, main window
config/settings.py	All constants, paths, and default values
config/user_prefs.json	Saved user preferences (auto-generated)
core/inventory.py	SQLite CRUD for inventory items

Path	Purpose
core/asset_manager.py	SQLite CRUD for assets, history, maintenance
core/barcode_tools.py	QR generation + scanner dialog
core/reporting.py	All report builders and export functions
core/prefs.py	User preference load/save/apply
core/backup.py	Database backup, restore, rotation
ui/main_window.py	Main application window — all 4 tabs
ui/item_editor.py	Full detail/edit dialog (double-click row)
data/inventrax.db	SQLite database (auto-created on first run)
logs/inventrax.log	Rotating application log
backups/	Timestamped database backup files
reports/	Exported report files
barcodes/	Generated QR code images

5. Deliverables

- Full Python source code for all modules listed in Section 4.2
- Single-file Windows executable (InventraX.exe) built via PyInstaller
- requirements.txt listing all Python dependencies with minimum version pins
- README.md covering installation, usage, keyboard shortcuts, backup/restore, and export
- Project Scope Document (this document)
- Terms and Conditions Document
- License Agreement

6. Assumptions & Dependencies

- The end user is running Windows 10 or Windows 11 (64-bit)
- Python 3.10 or newer is installed if running from source
- All listed pip dependencies are installable from PyPI
- Camera scanning requires a connected webcam and opencv-python + pyzbar installed
- PDF export requires reportlab installed
- The user has read/write access to the folder where the .exe or Main.py is located
- No internet connection is required to run the application

7. Known Limitations — v2.0.0

- Data is stored locally only — no cloud backup or remote access
- Single-user only — simultaneous access from multiple machines is not supported
- Camera scanning is not available if opencv-python or pyzbar are not installed
- PDF export requires reportlab to be installed separately
- The application has been tested on Windows; macOS and Linux are not officially supported in this release
- Inventory item names are used as the primary key — renaming an item creates a new record

8. Version History

Version	Date	Author	Changes
1.0.0	2025	Alejandro X. Solis	Initial release